Osmosis Experiments

Exam code: 0610 & 0970

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Updated on 23 September 2024

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Osmosis Experiments

Immersing plant cells in solutions of different concentrations

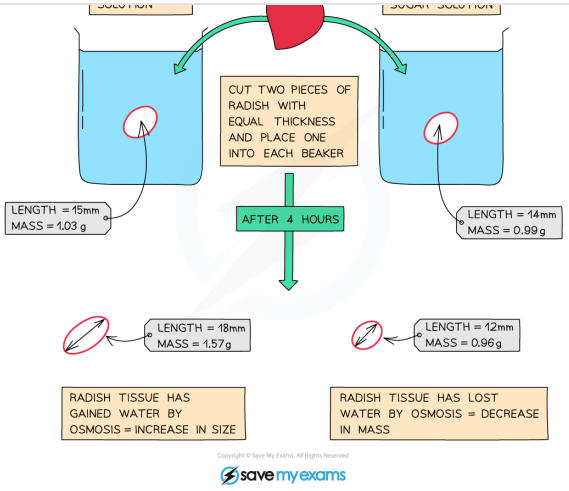
- The most common osmosis practical involves cutting **cylinders of root vegetables** such as potato or radish and placing them into **distilled water** and **sucrose solutions** of increasing concentration
- The cylinders are weighed before placing into the solutions
- They are left in the solutions for 20 - 30 minutes and then removed, dried to remove excess liquid and reweighed

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



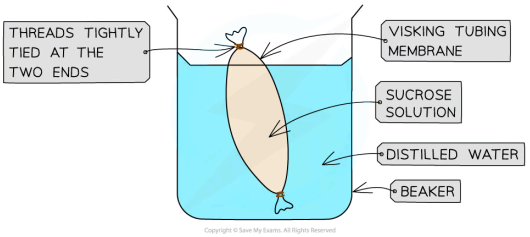
Potatoes are usually used in osmosis experiments to show how the concentration of a solution affects the movement of water, but radishes can be used too

- If the plant tissue gains mass:
  - Water must have moved into the plant tissue from the solution surrounding it by osmosis
  - The solution surrounding the tissue is more dilute than the plant tissue (which is more concentrated)
- If plant tissue loses mass:
  - Water must have moved out of the plant tissue into the solution surrounding it by osmosis
  - The solution surrounding the tissue is more concentrated than the plant tissue (which is more dilute)
- If there is no overall change in mass:
  - There has been no **net** movement of water as the concentration in both the plant tissue and the solution surrounding it must be equal
  - Remember that water will still be moving into and out of the plant tissue, but there wouldn't be any net movement in this case

Investigating osmosis using dialysis tubing

- **Dialysis tubing** (sometimes referred to as visking tubing) is a non-living **partially permeable** membrane made from cellulose
- Pores in this membrane are small enough to **prevent the passage of large molecules** (such as **sucrose**) but allow **smaller molecules** (such as **glucose and water**) to pass through by **diffusion** and **osmosis**
- This can be demonstrated by:
  - Filling a section of dialysis tubing with concentrated sucrose solution
  - Suspending the tubing in a boiling tube of water for a set period of time
  - Noting whether the water level outside the tubing decreases as water moves into the tubing via osmosis

a partially permeable membrane



An example setup of a dialysis tubing experiment

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